

Metrology for the second quantum revolution

ARTICLE INFO

Keywords:

Quantum technology
Quantum sensing
Validation
Metrology infrastructure

ABSTRACT

The ongoing second quantum revolution will lead to unprecedented technological capabilities opening up important scientific progress and business opportunities. The worldwide investments in quantum technology are important in the academic as well as in the industrial sector. However, these new technologies must also be duly characterised and calibrated with measurements traceable to the International System of Units, as it is the standard today for classical technologies. An accessible and fit for purpose testing infrastructure for quantum technology is thus clearly needed. In this paper we give an overview of our vision of a Swiss metrology competence centre for quantum technologies at the Swiss Federal Institute of Metrology METAS and highlight the present state of the recent developments.

1. Introduction

At the end of the 19th century, the understanding of physical laws was already well advanced. When Max Planck, as a young student, inquired about the prospects of studying physics around 1870, he was told that physics was a nearly completed science and, therefore, no more ground-breaking findings were to be expected. It is an ironic twist of history that it was the very same Max Planck who, at the turn of the century, provided the impetus for the first quantum revolution in physics with his explanation of black-body radiation through quantised energy states.

1.1. The advent of quantum mechanics

In the 1920s, a group of physicists, including Erwin Schrödinger, Werner Heisenberg, Paul Dirac and Wolfgang Pauli, formulated the foundations of quantum mechanics. This theory allowed, for the first time, to better understand the natural phenomena on the atomic scale. In the following decades, this new mechanics was developed to such an extent that the properties of atoms, molecules, solids and radiation could be precisely described.

The fact that the predictions of quantum mechanics are sometimes counter-intuitive and that the nature of the infinitely small exhibits a behaviour that differs from our everyday experience has also instilled doubts even in the most renowned physicists' minds. Albert Einstein, for example, struggled with the significance of quantum mechanical findings until the end of his life. Not only the stochastic behaviour, but also the "spooky" action at a distance for entangled quantum objects was difficult to accept for him. Only in the early 1980s, Alain Aspect, who was awarded the Nobel Prize in Physics in 2022, was able to prove by experimental means [1] that this entanglement, although difficult to understand, is indeed a physical reality.

1.2. The dawn of new possibilities

While the first quantum revolution led to an understanding of matter on its smallest scale and to indirect technical applications – for example in semiconductor technology – physicists and engineers learnt over time how to better control quantum properties and use them for technical purposes.

This led to the development of quantum cryptography, for example, which makes it possible to develop protocols that are information-theoretically secure. Thanks to features of quantum mechanics, namely the superposition principle and the no-cloning theorem for quantum states, it is possible to realise quantum key distribution (QKD) protocols which are in principle tap-proof. Nowadays, high secret key rates can be achieved with prepare-and-measure protocols by using weak coherent states – attenuated down to the single-photon level – and high-rate single-photon detectors [2]. Combined with classical encryption methods, messages can be secured at high rates which would already allow for quantum-secured video conferencing over metropolitan distances.

If not only a secret key has to be shared between two parties, but a quantum state needs to be transferred, another property of quantum mechanics comes into play: entanglement. This resource enables protocols such as quantum teleportation where a quantum state can be transferred from one location to another if the two involved parties share entanglement. The ultimate goal of quantum communication is to establish the quantum internet – a global quantum network where, for example, quantum states could be transferred between remote quantum computers [3].

Another application of quantum systems is the measurement of electromagnetic fields. Rydberg atoms, characterized by their outermost electron being loosely bound in a high quantum number orbit, exhibit heightened sensitivity to external electromagnetic fields [4]. In certain

This article is part of a special issue entitled: Supp: IMEKO 2024 published in Measurement: Sensors.

<https://doi.org/10.1016/j.measen.2025.101857>

Available online 17 February 2025

frequency ranges, this allows for more precise and sensitive measurements than with conventional methods.

The two examples mentioned above exploit technologies being part of the second quantum revolution and give an idea of the immense potential for development that we are currently experiencing and open up significant scientific advances and new business opportunities. As a result, global investment in quantum technology is high in both academia and industry. Given its strategic relevance, many countries have defined national quantum initiatives and strategies, fostering dynamic quantum ecosystems, also in Switzerland (see for instance: [5]). Swiss universities have a strong track record in fundamental research of quantum technologies, leading to a situation where, several Swiss SMEs and start-ups are well-positioned on the global market and are further exploring business opportunities. Furthermore, world leading quantum technology companies have established or strengthened branches in Switzerland to access this talent and technology pool of Swiss academies.

Several aspects of quantum technologies are generally considered as highly disruptive technologies, with the potential to bring fundamental changes to sensing, security and computing. Hence it is pivotal to create trust by standardisation and well-defined calibration procedures for these new quantum-based devices, the same way it is already common practice in classical devices; in particular, traceability to the International System of Units (SI) must be guaranteed. This creates the conditions for successful standardisation of quantum-based systems and will also strengthen confidence in quantum technologies.

Furthermore, well characterised quantum devices will provide a competitive advantage for quantum technology companies on the global market, bringing the precision and quality paradigm also into the quantum era. There is thus a clear need to ensure that the quantum technology ecosystems have access to a fit for purpose testing and measurement infrastructure. The need for such an infrastructure is also recognised by the European Commission [6] and activities have already been initiated.

2. Swiss competence centre for metrology in quantum

For the before mentioned reasons, METAS decided to further develop its expertise in the domain of quantum technologies and gradually develop a Swiss competence centre for metrology in quantum and ensure that also the Swiss quantum community has access to a suitable metrology infrastructure at METAS.

In metrology, quantum systems have been used as references for decades. Such systems represent immovable standards for various units based on natural constants. For instance, the electrical unit Ohm Ω for the electrical resistance is realised through the Quantum Hall effect, which can be traced back to the Planck constant and the electron charge [7], or the voltage standards for DC and AC defined by the same physical constants via the Josephson effect [8].

Quantum effects have also long been used to define the second: Atomic clocks, such as the "Fontaine Continue Suisse 2" (FoCS2) at METAS [9], define the international time scale UTC. They exploit measurements of the ground state transition frequency between quantised energy levels in thermally decelerated caesium atoms. Owing to these clocks, time and frequency can be measured with the highest precision amongst all physical quantities.

The experience acquired by these traditional applications of quantum properties in metrology, together with the strong quantum ecosystem in Switzerland, METAS is well positioned to develop and maintain an infrastructure for quantum metrology and to offer research capacities and relevant services. This will complement the national measurement base already operated by METAS.

On the other hand, due to the dense network and well-established role within the European Metrology community, METAS is excellently positioned to establish the needed coordination between Swiss and European efforts and help the Swiss industry to be compatible with the

European market. Moreover, as metrology is globally well interconnected, the European vision can also be brought in on a global level.

The need in metrological infrastructure can be categorised in three canonical topics of quantum technology: i) Quantum Communication, ii) Quantum Computing, and iii) Quantum Sensing. We propose to establish a set of measurement capabilities in each of these three branches and establish a Swiss competence centre for metrology in quantum. Finally, these technical capabilities should be complemented by further work streams including dissemination and educational activities.

In the following sections of this chapter, we identify and sketch the infrastructure needs and actions for the above mentioned four topics based on our current understanding of the ecosystem.

2.1. Quantum communication

Secure quantum communication infrastructures rely, among other components, on QKD systems. These systems are capable of establishing a secure key that the sender (receiver) can use to encode (decode) the message. As it is – due to the laws of quantum mechanics – inherently impossible to measure quantum states without influencing them, eavesdroppers will reveal themselves if they attempt to read the quantum key.

However, such components are only secure if the QKD systems fulfil strict specifications. Those specifications should be tested and the characteristics of the components assessed. Joining forces with other national metrology institutes in the framework of the European Metrology Programme for Innovation and Research (EMPIR) prepared the ground for such optical tests [10] enabling the detection of potential security holes within QKD systems.

We intend to broaden the platform for testing and characterisation of quantum communication equipment including the following capabilities:

- Characterisation of QKD components, i.e., large bandwidth spectral analysis and traceable measurements for transmission losses
- Calibration and characterisation of single-photon detectors and sources
- Platform for testing free space QKD links
- Quantum random number generator (QRNG) testing:
 - Assessing the randomness of QRNG by applying the random number test suites [11].
 - Physical characterisation and hardening tests of the quantum seed in function of temperature, external fields, vibrations, etc.
- Realisation and dissemination of traceable time and frequency signals to enable sub-nanosecond synchronicity in quantum communication networks. This activity is in synergy and continuity of existing activities in time and frequency dissemination of fibre optic networks [12].

2.2. Quantum Computing

Quantum computing systems push the parameter requirements both for supplying components as well as the quantum systems under test, such as superconducting technology, ultracold atoms, trapped ions, or semiconductor nanostructures to new limits. In order to fulfil the demanding requirements for high fidelity control and manipulation of qubits, the hardware has to follow strict specification in harsh environments. In order to support this sector, metrology can help assessing key physical parameters. In particular:

- Cryogenic scanning microwave microscopy for characterisation of superconducting and spin qubits
- Microwave to optical conversion: characterisation of electro-optical modulators to enable coherent state conservation

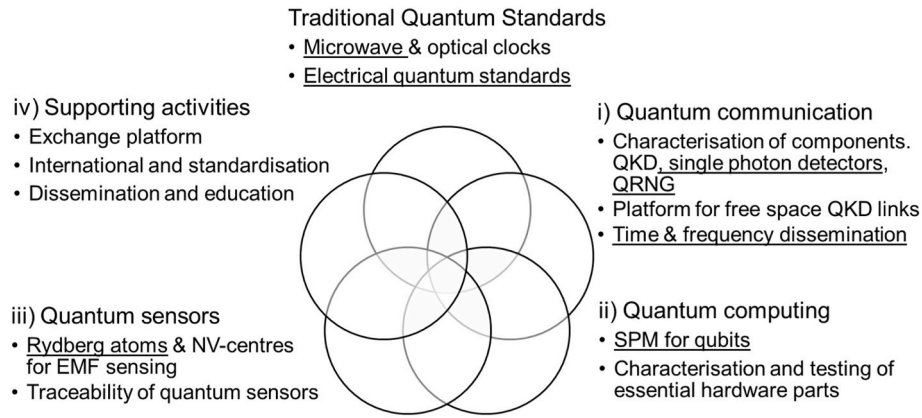


Fig. 1. Overview of the fields of activity within the planned Swiss Competence Centre for Metrology in Quantum envisaged by METAS (underlined items: active or planned fields, all others will have to be explored).

- Characterisation and testing of hardware parts – not necessarily using quantum properties – to enable the upcoming industrialisation of quantum computers

In this context it is worthwhile to mention, that for the first above mentioned bullet point, one project in the framework of the European Partnership on Metrology [13] has been accepted. This work will start during summer 2024 with the aim to bring scanning probe microscopy (SPM) to the cryogenic level. SPM will then be used to measure the two-level systems in superconducting qubits and in a parallel effort complex dielectric constants and currents.

2.3. Quantum sensing

Quantum sensing methods such as Rydberg atoms systems or nitrogen-vacancy (NV) centre microscopes are already close to market readiness. However, metrological characterisation and traceability have not yet been shown. Moreover, METAS is currently in the process of applying Rydberg atoms to electromagnetic field sensing for metrological applications. First feasibility studies in collaboration with the Basel Quantum Centre [14] have shown potential improvements compared to classical methods in certain frequency bands. Thus, we aim to establish the following measurement capabilities in the field of quantum sensors:

- Improve electric and magnetic field (EMF) measurements by using sensors based on Rydberg atoms or NV centres in diamond in different use cases
- Application of metrological traceability methods to quantum sensing methods using Rydberg atoms and NV centres.
- Develop measurement services for quantum EMF sensing devices

In a first step we will develop a method to probe electric fields above a coplanar waveguide using Rydberg atoms. This project in collaboration with Philipp Treutlein's quantum optics lab of the University of Basel [15] is funded by the European Partnership on Metrology [13].

2.4. Supporting activities

The development of know-how and capabilities is of key interest. In addition, in order to ensure that the needs of the quantum ecosystem are well captured, a close and dynamic collaboration and exchange with academy, industry and the international metrology community is paramount. Therefore, the technical capabilities will be enhanced and complemented by supporting activities such as communication and dissemination. The following points will be considered in this domain:

- Exchange platform with the Swiss quantum community to monitor the metrological needs and support standardisation efforts
- Develop concepts for sound integration of measurement uncertainties by using quantum methods
- Establish links and coordination to the global metrology community (METAS is founding member of the very active European Metrology Network for Quantum Technologies [16] which should facilitate this action point)
- Facilitate uptake of the methods in standardisation organisations (ISO, IEC, IUT and the related regional and national bodies)
- Dissemination and educational activities concerning metrology for quantum technologies within the Swiss quantum community

3. Present status and outlook

The concept of the Swiss competence centre for metrology in quantum at METAS as described in the previous section and depicted in Fig. 1 is quite recent. Nevertheless, a part of the above-listed activities is about to be started in collaboration with different partners, mainly within the framework of the European Partnership on Metrology. At the same time, an ambitious quantum technology program funded by the Swiss government has been launched and different calls for research and infrastructure development in Switzerland have been opened recently [17]. Besides the funding opportunities, this enables also the better coordination of activities within the Swiss quantum ecosystem and the positioning of exchange platforms to jointly advance quantum technology.

For the time being, the here described concept is our best bet to establish a metrology infrastructure for the ongoing quantum revolution. However, there will be other needs and challenges ahead of metrology in the quantum era in the years to come. We have thus to stay tuned to the international quantum community, integrate future trends and develop the metrology infrastructure accordingly to ensure a trusted adoption of this new technological possibilities.

Funding statement

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Acknowledgments

We would like to thank Andreas Masuhr, Head of the Swiss Quantum Initiative for the exchange about the Swiss quantum infrastructure as well as Patrick Maletinsky, Philipp Treutlein, Kai Wagner, and Tilman Zibold of the Basel Quantum Centre for their inputs on quantum sensing. Finally, we thank Dominik Husmann for his inputs, suggestions and the proofreading of the manuscript.

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